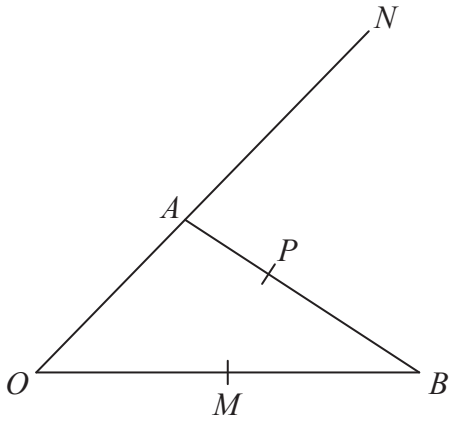
21



OAN, *OMB* and *APB* are straight lines.
$AN = 2OA$.
M is the midpoint of *OB*.

$\overrightarrow{OA} = \mathbf{a}$ $\overrightarrow{OB} = \mathbf{b}$

$\overrightarrow{AP} = k\overrightarrow{AB}$ where *k* is a scalar quantity.

Given that *MPN* is a straight line, find the value of *k*.

All AB, MN and MP expressions below are vectors.

AB = b − a

1 mark
Find the vector from A to B

ON = 3a; OM = ½b; MN = 3a − ½b

1 mark
Use AN = 2OA and the midpoint M

MP = MA + AP = (a − ½b) + k(b − a)

= (1 − k)a + (k − ½)b

1 mark
Find a second vector along MN

Since M, P, N are collinear, MP = t MN for a scalar t.

Compare coefficients: 1 − k = 3t; k − ½ = −½t.

1 mark
Use parallel vectors and match both coefficients

t = 1 − 2k, so 1 − k = 3(1 − 2k) = 3 − 6k.

5k = 2, so k = $\frac{2}{5}$

1 mark
Correct scalar value

$\frac{2}{5}$

(Total for Question 21 is 5 marks)